Ch. 24 HW #2 Solutions

Principles of Macroeconomics

January 7, 2026

Questions for Review

5. Consumption: A household buys a new blender.
   Investment: A company buys a new forklift.
   Government Purchases: The government buys a new tank.
   Net Exports: A person in Great Britain buys Microsoft Office

6. Unlike nominal GDP, real GDP isn't influenced by changes in the price level. Therefore if you compare real GDP across periods, it identifies the change in an economy's productive capacity with no distortions from changing prices. Productive capacity is a better measure of an economy's well-being because it determines the economy's ability to meet people's material needs.

7.

Year	GDP_{Nom}	GDP_{Real}	GDP Deflator
2023	$100(2) = 200$	$100(2) = 200$	$\frac{200}{200} \times 100 = 100$
2024	$200(3) = 600$	$200(2) = 400$	$\frac{600}{400} \times 100 = 150$

Problems & Applications

1. (a) Consumption – A personal refrigerator is household spending.
   (b) Investment – New housing is investment.
   (c) None – There's no spending on new production.
   (d) Consumption – A personal haircut is household spending.
   (e) Consumption and investment – Purchasing a personal car increases consumption, but decreasing inventory decreases investment by an offsetting amount.
   (f) Investment – A new car is a capital good for a rental car company because it allows them to produce more rental agreements.
   (g) Government purchases – Repaving roads requires government spending on current production.

(h) None – Transfer payments aren't spending on production.

(i) Consumption and net exports – Buying a foreign bottle of wine increases consumption, but buying an import decreases net exports by an offsetting amount.

(j) Investment – A factory is a capital good.

2.

Year	GDP_{Real}	GDP_{Nom}	GDP Deflator
1970	3,000	1,200	$\frac{1,200}{3,000}(100) = 40$
1980	5,000	$5,000\left(\frac{60}{100}\right) = 3,000$	60
1990	$\frac{6,000}{100} \times 100 = 6,000$	6,000	100
2000	8,000	8,000	100
2010	$\frac{15,000}{200} \times 100 = 7,500$	15,000	200
2020	10,000	$\left(\frac{300}{100}\right) = 30,000$	300
2030	20,000	50,000	$\frac{50,000}{20,000}(100) = 250$

3. GDP is spending on current production. Transfer payments aren't made in exchange for production, so they are excluded.